ZERO-HALLUCINATION AUDIT AGENT

Agentic Document Intelligence for Financial Data Extraction & Classification

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Every audit of a German company looks at its fixed assets: machines, buildings, cars, software. The client hands over its schedules (Anlagengitter, Anlagenverzeichnis, Anlagenspiegel) as PDFs, Excel and Word files, in a new layout every time. Today an assistant types them by hand into the auditor's standard working file, the *Arbeitsversion*. One wrong figure, and the audit stands on it. Build an AI agent that reads these documents, fills the Arbeitsversion with every figure traced to its source, and then checks its own work.

THE CHALLENGE

Build an agent that takes one hand-over document and gives back the filled Arbeitsversion: one row per asset on the Anlagenverzeichnis tab, one row per asset class on the Anlagengitter tab. Every schedule tells the same story in its own shape. For each asset or class it shows the cost at the start of the year, what was added, sold or moved, the cost at year end, the depreciation so far and for this year, and the book value now and a year ago. You do not need to know German accounting.

What matters is that every number you write is exactly the number printed in the document, at the place you say it is. If the PDF shows `19.425,96` in the Zugänge column of asset 210004, your output says 1,942,596 cents, field `zugang`, asset 210004, page 1, and a box around those characters. The model may read the page, understand the layout and decide which column means what and which row is a subtotal. It may not type a figure. Figures come from code: from the PDF text layer, the OCR output or the Excel cell, copied and parsed. If you are not sure, leave the value blank and say why. A blank costs nothing; a wrong value costs three right ones.

YOU GET

The empty Arbeitsversion

and its output schema: one row per asset on the Anlagenverzeichnis tab, one row per asset class on the Anlagengitter tab

Synthetic hand-overs

Text PDFs, scans and Excel, with answer keys. All names and figures are invented; the layouts copy real hand-overs

The scorer

The same scorer we use for the leaderboard

WHAT YOU HAND IN

- **Your code, as a GitHub repository**, with one command that runs your agent on a folder of documents and writes one output per document. We run that command on the hidden test set, so it must run without you. The repository is your main output. An interface is nice, but it does not make up for missing ideas.
- **For each document:** the filled Arbeitsversion (.xlsx) and a JSON file with one record per value: the value, the row and field it fills, its pointer, and a flag when you left it blank or found a problem.

WHICH DOCUMENTS

Three kinds of schedule, with the same content in different shapes. An **Anlagenspiegel** has one line per asset or per class and columns for cost, depreciation and book values. An **Anlagengitter**, often printed from SAP, has one block of three or four lines per class, sometimes in thousand euro with the minus sign after the number. An **Anlagenverzeichnis** is the asset register: one line per asset with its acquisition date, useful life and depreciation method. Each comes as Excel, as a PDF with a text layer, as a scanned PDF, or as a PDF that looks sharp but has no text to copy, because every letter is drawn as a shape. Some documents cover one year, some two; some have cover pages, blank pages or pages from another report.

RULES

01 Code has to write every figure. The model never writes a number.

02 Every value points to its source: page and box, or cell.

03 Checks need to be run to ensure accuracy and also completeness.

04 As few per-layout rules as possible: the final test uses layouts nobody has seen.

05 Read, never judge. Map the document exactly as printed; a client's broken sum is flagged, not fixed.

HOW WE JUDGE

+1	0	-3
Correct value	Flagged blank	Wrong value

Each amount the document shows counts as one value: on an asset line, a subtotal or a total. An empty cell in an amount column is a zero. A value whose pointer does not show it counts as wrong. A team with 900 correct values, 60 flagged blanks and 40 wrong ones scores 900 − 120 = 780; guessing does not pay. The final ranking runs on a hidden test set with layouts nobody has seen, so rules written for the sample layouts will not carry over.

RESOURCES

Use anything you like: open-source libraries, OCR engines, Claude or any other AI tool. We cannot share real client documents, sorry: everything you get is generated. The sample set has 72 documents in 10 layout families, about 4,000 rows and 36,000 values, each with its answer key. Unsure about an accounting term? Ask Claude; it will not make or break your project.